CSET485 AI AND SOCIETY

ASSIGNMENT 1

COMPAS Recidivism Risk Score Analysis

Submission details	
Student name	Riya Panwar
Roll number	S24BCAU0040
Course / section	AI and Society
Assigned dataset	COMPAS Recidivism Risk Score
Train-test random seed	40

Prepared using the analysis and outputs in the accompanying Jupyter notebook.

Executive Summary

This report evaluates two predictive models on the COMPAS recidivism dataset and examines how model specification, confident errors, and decision thresholds affect interpretation. The analysis uses 7,214 eligible records, with 5,049 training observations and 2,165 test observations. Linear Regression achieved a test R^2 of 0.4387; Logistic Regression achieved 67.53% accuracy at threshold 0.50. The two models disagreed on 166 test cases (7.67%) when each was thresholded as required. The stress tests found subgroup differences in the prior-count coefficient, substantial coefficient sensitivity when prior counts were omitted, 97 high-confidence errors, and fairness ratios that varied with threshold. These findings do not establish causal effects and do not justify standalone automated use in criminal-justice decisions.

1. Dataset and Preprocessing

The COMPAS Recidivism Risk Score dataset was loaded from the ProPublica COMPAS analysis repository. After target filtering, 7,214 observations were available. The selected predictors were age, sex, race, age category, juvenile felony count, juvenile misdemeanor count, other juvenile offense count, prior-offense count, and charge degree. The target for Linear Regression was decile_score; the target for Logistic Regression was is_recid.

The notebook checked the selected raw predictors and found no missing values in those columns. The preprocessing pipeline still includes median imputation for numeric predictors and most-frequent imputation for categorical predictors so it remains robust if missing values occur. Numerical features are standardized using StandardScaler, while categorical features are one-hot encoded with unknown categories ignored. The pipeline is fitted as part of model training to avoid using test-set statistics for preprocessing.

The data was split into 70% training and 30% test observations using random_state=40, producing 5,049 training rows and 2,165 test rows. Potential outliers were examined using the 1.5×IQR rule and retained unless a documented data error was identified. For the training data, the diagnostic flagged 21 age values (0.42%), 196 juvenile-felony counts (3.88%), 285 juvenile-misdemeanor counts (5.64%), 365 other juvenile-offense counts

(7.23%), and 306 prior-offense counts (6.06%). The juvenile-count variables have Q1=Q3=0, so their IQR is zero and positive counts are flagged by the rule. These values may be genuine records, not errors.

2. Part 1 — Model Comparison and Mismatch Analysis

Linear Regression was fitted to predict the continuous COMPAS decile score, while Logistic Regression predicted the binary recidivism outcome, is_recid. Both models used the same training and test split. Linear Regression achieved a test R^2 of 0.4387. Logistic Regression achieved 67.53% accuracy at the 0.50 probability threshold. Its confusion matrix against the actual outcome was [[806, 317], [386, 656]], with rows representing actual classes 0 and 1 and columns representing predicted classes 0 and 1.

For the required model-to-model comparison, Linear Regression predictions were converted to binary flags using the median predicted test score of 4.4071. Logistic Regression probabilities were converted to flags using 0.50. The agreement matrix, with rows for the Linear Regression flag and columns for the Logistic Regression flag, was [[1054, 28], [138, 945]]. There were 166 mismatches among 2,165 test observations, or 7.67%.

The models disagree because they estimate different targets and optimize different objectives. Linear Regression minimizes squared error for a numerical score, whereas Logistic Regression estimates the probability of a binary outcome. Their thresholds also operate on different scales. The agreement matrix measures whether the models flag the same people; it does not show which model is correct. The Logistic Regression confusion matrix provides a separate comparison with the observed binary outcome. These results show why model choice and threshold selection matter in a high-stakes setting. Accuracy and agreement should be considered alongside error types, subgroup performance, uncertainty, and the consequences of an incorrect classification. This distinction is important.

Table 1. Logistic Regression confusion matrix against actual is_recid (rows = actual class; columns = predicted class).

Actual / predicted	Predicted 0	Predicted 1
Actual 0	806	317
Actual 1	386	656

Table 2. Agreement matrix between the model-based flags (rows = Linear Regression flag; columns = Logistic Regression flag).

OLS flag / Logistic flag	Flag 0	Flag 1
Flag 0	1,054	28
Flag 1	138	945

3. Part 2 — The Trap: Four Stress Tests

3.1 Simpson's Paradox Hunt

The subgroup analysis examined the coefficient for priors_count in a full logistic regression model and separate models for African-American and Caucasian individuals. The full-sample coefficient was 0.1517 (95% CI: 0.1372–0.1662), the African-American subgroup coefficient was 0.1432 (95% CI: 0.1249–0.1614), and the Caucasian subgroup coefficient was 0.1766 (95% CI: 0.1474–0.2058). All three estimates were positive, so the results do not show a reversal of sign. Instead, the estimated association was numerically stronger in the Caucasian subgroup.

The interaction model reported a coefficient of 0.0465 for the Caucasian-by-priors_count interaction (95% CI: 0.0139–0.0791, p=0.0051). This suggests that the association between prior counts and recidivism differs by group under the model specification. The coefficient is on the log-odds scale; it is not a percentage-point increase in probability. Exponentiating the interaction coefficient gives an odds multiplier of approximately 1.048 per additional prior count relative to the reference group.

This analysis should be described as a subgroup-effect comparison, not as proof of a Simpson's Paradox reversal. The models are observational, and their coefficients do not establish that race causes recidivism. Differences may reflect unmeasured variables, data collection practices, or other structural factors. If policymakers looked only at the fullsample coefficient, they might miss differences in the fitted relationship across groups. The responsible conclusion is to report subgroup estimates and uncertainty, investigate plausible explanations, and avoid assuming that one overall coefficient describes every group equally well. Such differences merit careful review.

Model	N	Coefficient	95% CI	p-value
Full dataset	7,214	0.1517	0.1372–0.1662	<0.001
African-American	3,696	0.1432	0.1249–0.1614	<0.001
Caucasian	2,454	0.1766	0.1474–0.2058	<0.001
Caucasian × priors_count interaction	—	0.0465	0.0139–0.0791	0.0051

3.2 Omitted-Variable Bias

The omitted-variable experiment compared a full logistic regression model containing priors_count with a reduced model that excluded it. Four other coefficients were tracked. Age changed from −0.0445 to −0.0278, a 37.49% change relative to the absolute full-model coefficient. juv_fel_count changed from 0.1976 to 0.4887, a 147.35% change. juv_misd_count changed from 0.0486 to 0.3871, a 695.71% change. juv_other_count changed from 0.1953 to 0.2974, a 52.28% change. The largest percentage shift was for juv_misd_count.

The percentage change was calculated as the absolute difference between reduced and full coefficients divided by the absolute full-model coefficient, multiplied by 100. The large percentage for juvenile misdemeanors is partly influenced by the small starting coefficient. The absolute change and confidence intervals should also be considered. These results show that estimates can be sensitive to model specification when a plausible predictor is omitted.

This experiment does not prove that the reduced-model coefficients have causal bias of exactly these magnitudes. Logistic-regression coefficients can change due to confounding and the non-collapsibility of odds ratios. Nevertheless, the large changes are a warning against interpreting a single fitted model as definitive. In practice, analysts should use domain knowledge to identify plausible omitted factors, compare alternative specifications, validate models on data from other places or time periods, and document coefficient stability. When estimates shift substantially, conclusions should be presented with caution and the omitted-variable question should be investigated further. The reported percentages describe coefficient changes under these two specifications; they should not be interpreted as a direct percentage change in an individual's probability of recidivism.

Predictor	Full coefficient	Reduced coefficient	Percentage change
age	−0.0445	−0.0278	37.49%
juv_fel_count	0.1976	0.4887	147.35%
juv_misd_count	0.0486	0.3871	695.71%
juv_other_count	0.1953	0.2974	52.28%

3.3 Adversarial Subset Construction

The adversarial analysis defined a confident error as a false positive with predicted probability at least 0.75 or a false negative with predicted probability at most 0.25. The notebook identified 97 such cases among 2,165 test observations, equivalent to 4.48%. A reproducible random sample of 50 cases was selected for closer inspection.

The sample's mean age was 47.64 years, compared with 34.88 years in the full test set. Women accounted for 16.00% of the sample compared with 19.08% of the test set. African-American individuals accounted for 48.00% compared with 49.61% of the full test set, while Caucasian individuals accounted for 40.00% compared with 34.50%. Prior-offense counts of five or more appeared in 44.00% of the sample, compared with 26.14% of the full test set.

These are descriptive differences and do not show that age, race, sex, or prior history caused the errors. The subset includes both false positives and false negatives, which may have different explanations. Logistic Regression imposes

a linear relationship between predictors and log-odds unless nonlinear terms or interactions are added. It may be confident when an individual's feature combination resembles patterns in the training data, even when relevant context is missing or the observed label is imperfect. A probability is a model estimate, not a guarantee.

A useful next step is to analyze false positives and false negatives separately, inspect their feature distributions and predicted probabilities, and test whether justified interaction terms or nonlinear features improve performance on a separate validation set. The results support further audit and caution, not automatic conclusions about any demographic group.

Measure	Full test set	Sample of 50
Number of observations	2,165	50 of 97 confident errors
Mean age	34.88	47.64
Female	19.08%	16.00%
African-American	49.61%	48.00%
Caucasian	34.50%	40.00%
Prior count ≥ 5	26.14%	44.00%

3.4 Fairness Trade-Off Analysis

Fairness was evaluated by comparing African-American individuals (Group A) with Caucasian individuals (Group B) across thresholds from 0.05 to 0.95 in increments of 0.10. Demographic Parity (DP) is the ratio of positive prediction rates; Equalized Odds (EO) is defined here as the minimum of the TPR ratio and FPR ratio; Predictive Parity (PP) is the ratio of precision among positive predictions. Under these ratio definitions, 1.0 represents equality between the groups. The distance from 1.0 is therefore useful for describing how far a metric is from its ideal value, but it is not a complete measure of fairness.

At threshold 0.35, DP was 1.3942 and EO was 1.2288. Their absolute distances from 1.0 were 0.3942 and 0.2288. At threshold 0.55, DP increased to 2.4103 and EO to 1.8728, giving distances of 1.4103 and 0.8728. The DP deviation from 1 increased by approximately 257.8%, calculated as (1.4103 − 0.3942) / 0.3942 × 100. The EO deviation increased by approximately 281.5%, calculated as (0.8728 − 0.2288) / 0.2288 × 100. Both metrics moved farther from their ideal value at the higher threshold.

Other metrics improved at 0.55. Accuracy rose from 62.73% at threshold 0.35 to 66.97%, an increase of 4.24 percentage points. PP moved closer to 1.0, from 1.1662 to 1.0280. However, the positive prediction rates for AfricanAmerican and Caucasian individuals changed from 83.99% and 60.24% at threshold 0.35 to 51.30% and 21.29% at threshold 0.55. Thus, the gap in positive prediction rates widened even though fewer people were classified positive in both groups.

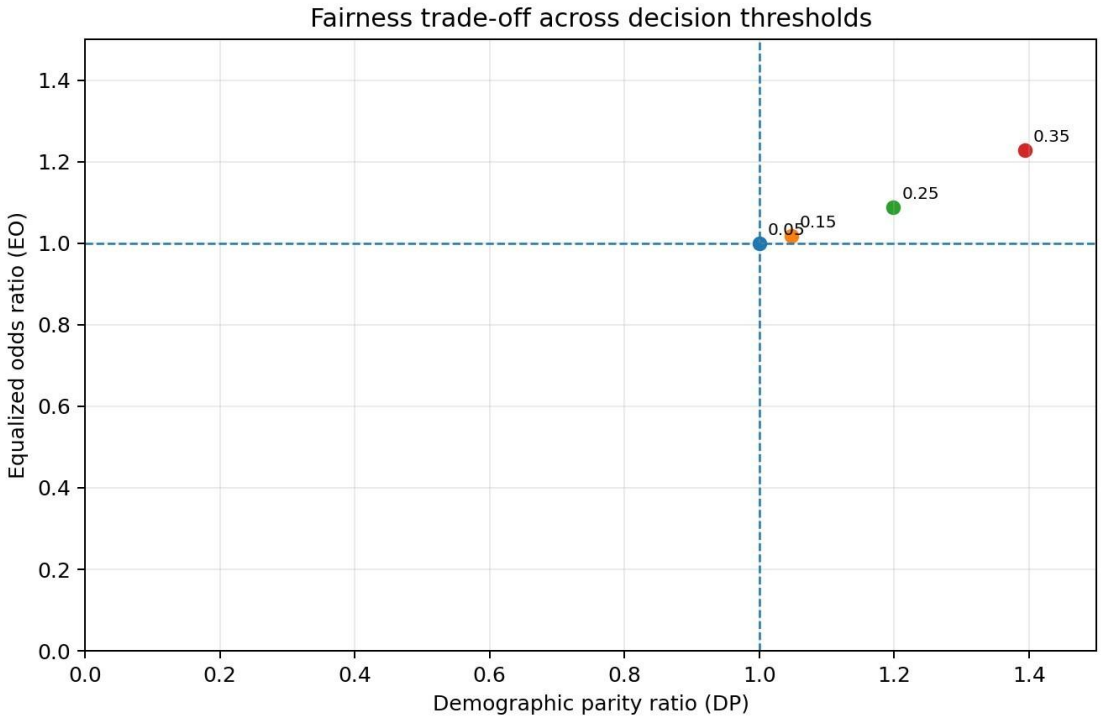
This comparison illustrates the observed trade-off: threshold 0.55 had higher accuracy and a PP ratio closer to 1, but its DP and EO ratios were farther from 1.0. Threshold 0.35 had DP and EO ratios closer to equality, but lower accuracy and a PP ratio farther from 1.0. The notebook's full threshold sweep found no pair in which DP moved closer to parity while EO moved farther from parity; therefore, the requested DP-improvement/EO-deterioration trade-off cost is not defined for this sweep. The comparison instead shows that improving accuracy and PP can coincide with worse DP and EO. The trade-off cost is not a single universal fairness score; it depends on which metric is prioritized and on the harms of false positives and false negatives. I would treat 0.35 as a candidate for further review rather than claim it is definitively fair. Before any high-stakes use, the choice should be evaluated using group-specific error rates, uncertainty intervals, external validation, consultation with affected communities, and meaningful human review.

Metric	Threshold 0.35	Threshold 0.55	Interpretation
DP	1.3942	2.4103	Moves farther from 1.0
DP distance from 1	0.3942	1.4103	Deviation increases by 257.8%
EO	1.2288	1.8728	Moves farther from 1.0

EO distance from 1	0.2288	0.8728	Deviation increases by 281.5%
PP	1.1662	1.0280	Moves closer to 1.0
Accuracy	62.73%	66.97%	Increases by 4.24 percentage points

Threshold	DP	EO	PP	Accuracy	Group A / B positive rate
0.05	1.0000	1.0000	1.3230	48.13%	100.0% / 100.0%
0.15	1.0472	1.0192	1.2876	49.79%	99.3% / 94.8%
0.25	1.1990	1.0884	1.2010	55.10%	95.3% / 79.5%
0.35	1.3942	1.2288	1.1662	62.73%	84.0% / 60.2%
0.45	1.8556	1.6461	1.1737	66.51%	68.8% / 37.1%
0.55	2.4103	1.8728	1.0280	66.97%	51.3% / 21.3%
0.65	2.8815	2.0700	0.9504	63.51%	29.7% / 10.3%
0.75	3.9558	2.9011	0.9703	58.66%	17.0% / 4.3%
0.85	4.6369	3.3564	0.9577	55.80%	9.3% / 2.0%
0.95	4.8687	N/A	0.7619	52.56%	2.0% / 0.4%

Figure 1. DP versus EO ratios for thresholds that fall within the assignment's displayed 0–1.5 axis range. Thresholds outside this range remain represented in the table.



4. Part 3 — Descriptive Analysis Questions

Question 1. Why do the models flag different individuals?

Linear Regression was fitted to predict the continuous COMPAS decile score, while Logistic Regression predicted the binary recidivism outcome, is_recid. Both models used the same training and test split. Linear Regression achieved a test R^2 of 0.4387. Logistic Regression achieved 67.53% accuracy at the 0.50 probability threshold. Its confusion matrix against the actual outcome was [[806, 317], [386, 656]], with rows representing actual classes 0 and 1 and columns representing predicted classes 0 and 1.

For the required model-to-model comparison, Linear Regression predictions were converted to binary flags using the median predicted test score of 4.4071. Logistic Regression probabilities were converted to flags using 0.50. The agreement matrix, with rows for the Linear Regression flag and columns for the Logistic Regression flag, was [[1054, 28], [138, 945]]. There were 166 mismatches among 2,165 test observations, or 7.67%.

The models disagree because they estimate different targets and optimize different objectives. Linear Regression minimizes squared error for a numerical score, whereas Logistic Regression estimates the probability of a binary outcome. Their thresholds also operate on different scales. The agreement matrix measures whether the models flag the same people; it does not show which model is correct. The Logistic Regression confusion matrix provides a separate comparison with the observed binary outcome. These results show why model choice and threshold selection matter in a high-stakes setting. Accuracy and agreement should be considered alongside error types, subgroup performance, uncertainty, and the consequences of an incorrect classification. This distinction is important.

Question 2. What does the subgroup analysis show?

The subgroup analysis examined the coefficient for priors_count in a full logistic regression model and separate models for African-American and Caucasian individuals. The full-sample coefficient was 0.1517 (95% CI: 0.1372–0.1662), the African-American subgroup coefficient was 0.1432 (95% CI: 0.1249–0.1614), and the Caucasian subgroup coefficient was 0.1766 (95% CI: 0.1474–0.2058). All three estimates were positive, so the results do not show a reversal of sign. Instead, the estimated association was numerically stronger in the Caucasian subgroup.

The interaction model reported a coefficient of 0.0465 for the Caucasian-by-priors_count interaction (95% CI: 0.0139–0.0791, p=0.0051). This suggests that the association between prior counts and recidivism differs by group under the model specification. The coefficient is on the log-odds scale; it is not a percentage-point increase in probability. Exponentiating the interaction coefficient gives an odds multiplier of approximately 1.048 per additional prior count relative to the reference group.

This analysis should be described as a subgroup-effect comparison, not as proof of a Simpson's Paradox reversal. The models are observational, and their coefficients do not establish that race causes recidivism. Differences may reflect unmeasured variables, data collection practices, or other structural factors. If policymakers looked only at the fullsample coefficient, they might miss differences in the fitted relationship across groups. The responsible conclusion is to report subgroup estimates and uncertainty, investigate plausible explanations, and avoid assuming that one overall coefficient describes every group equally well. Such differences merit careful review.

Question 3. What changed when priors_count was omitted?

The omitted-variable experiment compared a full logistic regression model containing priors_count with a reduced model that excluded it. Four other coefficients were tracked. Age changed from −0.0445 to −0.0278, a 37.49% change relative to the absolute full-model coefficient. juv_fel_count changed from 0.1976 to 0.4887, a 147.35% change. juv_misd_count changed from 0.0486 to 0.3871, a 695.71% change. juv_other_count changed from 0.1953 to 0.2974, a 52.28% change. The largest percentage shift was for juv_misd_count.

The percentage change was calculated as the absolute difference between reduced and full coefficients divided by the absolute full-model coefficient, multiplied by 100. The large percentage for juvenile misdemeanors is partly

influenced by the small starting coefficient. The absolute change and confidence intervals should also be considered. These results show that estimates can be sensitive to model specification when a plausible predictor is omitted.

This experiment does not prove that the reduced-model coefficients have causal bias of exactly these magnitudes. Logistic-regression coefficients can change due to confounding and the non-collapsibility of odds ratios. Nevertheless, the large changes are a warning against interpreting a single fitted model as definitive. In practice, analysts should use domain knowledge to identify plausible omitted factors, compare alternative specifications, validate models on data from other places or time periods, and document coefficient stability. When estimates shift substantially, conclusions should be presented with caution and the omitted-variable question should be investigated further. The reported percentages describe coefficient changes under these two specifications; they should not be interpreted as a direct percentage change in an individual's probability of recidivism.

Question 4. How would you detect a hidden confound?

A hidden confound cannot be ruled out merely because a regression model includes several predictors. I would use a concrete validation process. First, consult criminal-justice practitioners, researchers, and affected communities to map how risk scores are used and identify plausible missing factors, such as differences in policing exposure, access to legal support, or local supervision practices. These are hypotheses to investigate rather than assumptions about individuals. Second, test the model on data from a different jurisdiction or later time period, using consistent outcome definitions. Compare coefficient stability, calibration, false-positive rates, false-negative rates, and subgroup performance. A large change across settings may indicate that the model has learned context-specific patterns.

Third, conduct sensitivity analysis by fitting justified alternative specifications, checking interactions and nonlinear relationships, and documenting which estimates change. Residual and error analysis can reveal systematic patterns, but cannot prove that no hidden confound exists. If a causal claim is essential, a qualified research team might consider a natural experiment or a carefully justified instrumental-variable design; an instrument must satisfy strong assumptions and should not be selected simply because it changes the result.

The COMPAS experiment provides a reason to perform this work: removing priors_count changed the juv_misd_count coefficient from 0.0486 to 0.3871. That is a substantial specification change, although it does not alone establish a causal bias of that magnitude. The practical response is to investigate, validate independently, and report uncertainty. A transparent model card should describe data provenance, intended use, known limitations, subgroup results, and unresolved questions. The aim is not to claim that all confounding has been eliminated, but to make the remaining uncertainty visible before the model is used in consequential decisions.

Question 5. What characterizes the adversarial subset?

The notebook identified 97 confident errors, or 4.48% of the test set, and sampled 50 of those cases. In the sample, 44.00% had at least five prior offenses, compared with 26.14% of the full test set. The sample was also older on average: 47.64 years compared with 34.88 years. Women represented 16.00% of the sample compared with 19.08% of the test set. The African-American share was 48.00%, close to the test-set share of 49.61%, while the Caucasian share was 40.00% compared with 34.50%.

The observed characteristics are useful for directing further investigation, but they do not explain every error or establish that demographic attributes cause model failure. The subset includes confident false positives and confident false negatives. Those should be separated because a false positive and a false negative can arise from different patterns. Logistic Regression fits a linear decision boundary in the transformed feature space. It may fail when relationships are nonlinear, when interactions are important, when training examples do not adequately represent a feature combination, or when the observed outcome is noisy. Missing contextual variables can also limit predictions.

The model's confidence is conditional on its learned parameters and input features; it is not a guarantee that the prediction is correct for a particular person. A stronger audit would examine the probabilities and feature values of false positives and false negatives separately, check whether errors cluster by age or prior-count ranges, and evaluate alternative model specifications on a held-out validation set. The result supports targeted review and careful monitoring rather than the assumption that any individual with a particular profile is inherently likely to recidivate.

Question 6. What does the fairness trade-off mean for people?

The fairness trade-off has practical consequences because a threshold can change a person's classification even when the person's circumstances have not changed. To illustrate, consider a hypothetical African-American individual with predicted probability 0.40 and a hypothetical Caucasian individual with probability 0.30. At threshold 0.35, the first would be classified positive and the second negative; at threshold 0.55, both would be classified negative. This example is illustrative, not a claim about two identified rows from the test set. Whether either classification is correct depends on the observed outcome and the limitations of the label.

The group-level results show why the threshold choice matters. At 0.35, the positive prediction rates were 83.99% for African-American individuals and 60.24% for Caucasian individuals, with DP=1.3942. At 0.55, the rates were

51.30% and 21.29%, and DP rose to 2.4103. EO also rose from 1.2288 to 1.8728, while PP moved closer to equality from 1.1662 to 1.0280. Thus, a change that improves one metric can worsen another.

For a real person, a false positive may lead to a more restrictive decision, while a false negative may lead a decisionmaker to underestimate risk. The relative harm depends on how the score is used. A responsible threshold decision should therefore consider the purpose of the model, the reliability of outcome labels, group-specific error rates, and the consequences of mistakes. The model should not make consequential decisions automatically. Policymakers should explain the selected fairness objective, consult affected communities, and provide meaningful human review and an appeal process.

Question 7. Is threshold 0.35 defensible?

Threshold 0.35 is a defensible candidate for discussion, but the results do not establish it as the uniquely correct threshold. At 0.35, the notebook reports accuracy of 62.73%, DP=1.3942, EO=1.2288, and PP=1.1662. The positive prediction rates were 83.99% for African-American individuals and 60.24% for Caucasian individuals. These values show that the groups do not have equal positive prediction rates, but the DP ratio is closer to 1 than it is at threshold 0.55, where DP rises to 2.4103 and EO to 1.8728.

A critic could reasonably argue that threshold 0.55 has higher test accuracy, 66.97%, and a PP ratio closer to 1, at 1.0280. That is a valid counterargument. On the other hand, selecting the threshold solely by accuracy would ignore the larger DP and EO ratios and the possible consequences of different false-positive and false-negative rates. The decision is therefore partly a policy judgment about which errors matter most, rather than a purely technical optimization.

Before choosing threshold 0.35 for real use, I would examine uncertainty intervals for group metrics, compare subgroup TPR and FPR directly, test calibration, and evaluate performance on data from a different time period or jurisdiction. I would also ask affected communities and decision-makers to state which harms they are prioritizing. The threshold is defensible only as a transparent option for further evaluation, not as proof that the model is fair or suitable for automated criminal-justice decisions.

Question 8. Which fairness objective should be prioritized?

If policymakers must choose a primary metric for this COMPAS analysis, I would prioritize scrutiny of equalized odds and the consequences of false positives and false negatives, while still reporting demographic parity and predictive parity. Equalized odds concerns differences in true-positive and false-positive rates, which are directly related to whether the model makes different types of errors across groups. It is not, however, a complete definition of fairness.

At threshold 0.35, the notebook reports EO=1.2288 and DP=1.3942. At threshold 0.55, EO increases to 1.8728 and DP to 2.4103, while PP moves closer to 1 at 1.0280 and accuracy increases to 66.97%. This shows that a threshold with better accuracy and predictive parity can have worse group parity under the other metrics. Under the ratio definitions used here, 1.0 represents equality; the further a ratio moves from 1, the larger the difference.

Prioritizing equalized odds may mean accepting unequal positive prediction rates or predictive parity, and those costs must be acknowledged explicitly. Conversely, prioritizing demographic parity alone can obscure differences in error rates and can encourage a threshold that classifies many people positive even when model accuracy is poor. At threshold 0.05, DP equals 1.0 because both groups have a 100% positive prediction rate, but overall accuracy is only 48.13%.

I would not recommend automated deployment based on one metric. The decision should reflect the intended use, the validity of the outcome label, the relative harms of errors, and consultation with affected communities. All three metrics, group-specific error rates, and uncertainty should be reported. The goal is a transparent and accountable decision, not the claim that one fairness ratio proves the system fair.

5. Part 4 — Reflection: Should COMPAS Be Deployed?

I would not recommend deploying COMPAS as a standalone decision-maker. At most, its predictions could be considered within a tightly controlled, independently validated process with human review, transparent limitations, and a meaningful way for affected people to challenge decisions.

The first reason one number is insufficient is that overall accuracy hides the kinds of errors a model makes. Logistic Regression achieved 67.53% test accuracy, but at the 0.50 threshold its confusion matrix contained 317 false positives and 386 false negatives. The model-to-model comparison also found 166 mismatches, or 7.67% of the test set, between the Linear Regression-based flags and Logistic Regression flags. Accuracy alone cannot tell a policymaker who was wrongly flagged, whether error rates differ across groups, or what the consequences of an error may be. It also does not establish that the target label fully captures the concept that policymakers intend to measure.

Second, fairness results depend on both the threshold and the metric. At threshold 0.35, DP was 1.3942, EO was 1.2288, PP was 1.1662, and accuracy was 62.73%. At threshold 0.55, accuracy rose to 66.97% and PP moved closer to 1 at 1.0280, but DP increased to 2.4103 and EO to 1.8728. A threshold is not simply a technical fact; it reflects a choice about which errors and group differences decision-makers are willing to tolerate. No single ratio captures every relevant idea of fairness, and choosing one metric may leave another group-level concern unresolved.

Third, the model can be confidently wrong. The adversarial analysis found 97 high-confidence errors, representing 4.48% of the test set. In the sample of 50 errors, 44.00% had at least five prior offenses, compared with 26.14% of the full test set. This is a reason to investigate the error patterns, not a reason to treat prior history as destiny. Confidence is a model output, not a guarantee of correctness. False positives and false negatives may also arise for different reasons and should be audited separately.

Fourth, estimates were sensitive to model specification. Removing priors_count changed the juv_misd_count coefficient from 0.0486 to 0.3871, a reported 695.71% change relative to the full-model coefficient. This warns that conclusions may depend on which predictors are included. It does not alone prove a causal effect of that magnitude, but it supports further sensitivity testing and external validation. The subgroup analysis also found different priors_count coefficients across groups, without a sign reversal, reinforcing the need to avoid relying on one overall estimate.

Before any use, I would require independent validation, subgroup error analysis, a documented threshold, monitoring over time, and a human review and appeal process. The data, outcome definitions, and limitations should be explained to decision-makers and affected communities. The broader lesson is that responsible AI requires a transparent account of uncertainty and trade-offs, not a single score that hides them.

Deployment should remain conditional on evidence that the system is valid for its intended setting.

6. Limitations and Reproducibility

The results describe one train-test split and depend on the selected features, target definition, group comparison, and threshold. Regression coefficients are associations rather than causal effects. Fairness ratios can be undefined when their denominator is zero; these values should be reported as unavailable rather than interpreted as zero or as evidence of perfect fairness. The notebook should be run from the first cell to the last to regenerate the tables and figures. Any deployment decision would require independent validation, documented governance, human oversight, and continued monitoring.